

Mixed Noise Removal via Laplacian Scale Mixture Modeling and Nonlocal Low-rank Approximation

Tao Huang, Weisheng Dong, *Member, IEEE*, Xuemei Xie, Guangming Shi, *Senior member, IEEE*, and Xiang Bai

Abstract—Recovering the image corrupted by additive white Gaussian noise (AWGN) and impulse noise (IN) is a challenging problem due to its difficulties in accurate modeling of the distributions of the mixture noise. Many efforts have been made to first detect the locations of the impulse noise and then recover the clean image with image inpainting techniques from an incomplete image corrupted by AWGN. However, it is quite challenging to accurately detect the locations of the impulse noise when the mixture noise is strong. In this paper, we propose an effective mixture noise removal method based on Laplacian scale mixture (LSM) modeling and nonlocal low-rank regularization. The impulse noise is modeled with LSM distributions, and both the hidden scale parameters and the impulse noise are jointly estimated to adaptively characterize the real noise. To exploit the nonlocal self-similarity and low-rank nature of natural image, a nonlocal low-rank regularization is adopted to regularize the denoising process. Experimental results on synthetic noisy images show that the proposed method outperforms existing mixture noise removal methods.

Index Terms—mixed noise removal, Low-rank, Laplacian scale mixture, alternative minimization.

I. INTRODUCTION

Recovering a clean image from its corrupted version is one of the most basic tasks in image processing, and it has been extensively studied in the past decades [1]–[14]. Generally, two types of noise are often encountered during image acquisition and transmission, i.e., the additive white Gaussian noise (AWGN) and the impulse noise (IN). The AWGN is commonly introduced during image acquisition, while IN is often generated by the damaged pixels in camera sensors and transmission errors [15]. In practice, images are often corrupted by the mixture of the two types of noise, making the denoising problem even difficult. To faithfully recover the clean images with sharp edges and fine image details, the selection of the noise model and the prior knowledge of natural images are of great importance.

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T. Huang and W. Dong are with the State Key Laboratory on Integrated Services Networks (ISN), School of Electronic Engineering, Xidian University, Xi'an, 710071, China (e-mail: htt1001@163.com; ws-dong@mail.xidian.edu.cn)

X. Xie and G. Shi are with School of Electronic Engineering, Xidian University, Xi'an, 710071, China (e-mail: xmxie@mail.xidian.edu.cn; gmshi@xidian.edu.cn)

X. Bai is with the Department of Electronics and Information Engineering, Huazhong University of Science and Technology, Wuhan, 430074, China (e-mail: xbai@hust.edu.cn).

For images corrupted with Gaussian noise only, the Gaussian model, corresponding to the ℓ_2 -norm of the data fidelity term, perfectly fits the noise distributions. With an appropriate regularizer incorporating the prior knowledge of natural images (e.g., the piecewise smoothness [16], the sparsity prior [2]–[5]), the Gaussian noise can be effectively removed by solving the following optimization problem

$$\hat{x} = \underset{x}{\operatorname{argmin}} \|y - x\|_2 + \lambda J(x), \quad (1)$$

where $y = x + w$, $y \in \mathbb{R}^N$ and $x \in \mathbb{R}^N$ denotes the noisy image and clean image, respectively, and $J(x)$ denotes the regularization term. Classic regularizers based on smoothness prior includes the quadratic Tikhonov regularization [17] and the Total Variation (TV) regularization [16], [18]. In the past decade, the sparsity-based regularizer has been proven to be a powerful and popular tool for image denoising, which can be expressed as

$$\min_{x, \alpha_i} \|y - x\|_2 + \lambda \sum_i \|R_i x - D \alpha_i\|_2^2 + \eta \sum_i \psi(\alpha_i), \quad (2)$$

where $R_i \in \mathbb{R}^{n \times N}$ denotes the matrix extracting an image patch of size $\sqrt{n} \times \sqrt{n}$ centered at the pixel position i , $D \in \mathbb{R}^{n \times K}$ denotes the dictionary, and $\psi(\alpha_i)$ denotes the sparsity penalty on the sparse codes $\alpha_i \in \mathbb{R}^K$. The rigorous sparse penalty is the ℓ_0 -norm that counts the number of non-zeros. As the ℓ_0 -minimization is a combinatorial optimization problem, its convex envelope, i.e., ℓ_1 -norm is widely used. There have been numerous efforts in developing fast and effective optimization algorithms for ℓ_1 -minimization problems [19]–[21] which further boost the denoising performance. As the ℓ_0 or ℓ_1 norms assumes the independencies between the sparse codes, recent advanced sparse regularizers also exploit the correlations between the sparse coefficients [3]–[6]. By exploiting the correlations between the sparse coefficients, the sparse coding becomes more robust and accurate, leading to improved denoising performance. In addition to the sparsity-based denoising methods, the low-rank based denoising methods have shown even more effective denoising performance [7], [8] based on the assumption that the matrixes formed by the similar patches are low-rank.

For real image denoising applications, the noise parameter, i.e., the noise variance is unknown and has to be estimated. There have been extensive studies on noise parameter estimation in the past decades. A widely used noise estimation method is the mean absolute deviation (MAD) method in the HH wavelet subband [22]. However, despite its simplicity, the estimation by the MAD method is often inaccurate due

to influence of image details in HH subband. Filtering-based methods [23], [24] have also been developed. In these methods, the noisy image is first filtered with a high-pass filter and the noise parameter is then estimated from the filtered image, which is assumed to be noise. However, this assumption is not always true, leading to biased estimates. Patch-based methods [25] attack this problem by first extracting homogenous image patches from the noisy image and then computing the noise parameter from the selected homogenous patches. However, it is often difficult to select the homogenous patches from a noisy image. Recently, PCA-based methods have been proposed in [26]–[28], where the noise variances are estimated from the eigenvalues of the image patches, showing state-of-the-art performances. An advanced version of the PCA-based method with theoretical guarantee has also been reported in [29]. With the estimated noise parameter, the above mentioned Gaussian noise denoising methods can be applied. Though excellent denoising performance have been achieved, these methods based on the AWGN model are sensitive to the outliers and non-Gaussian noise, such as the impulse noise.

In general, there are two typical IN, i.e., the salt-and-pepper IN (SPIN) and random valued IN (RVIN). Conventional IN removal methods use nonlinear filters, such as the median filters [30]. The median filtering is effective in removing IN at the expense of blurring image local structures. To better preserve image edges, a set of improved median filtering techniques have been proposed, e.g., weighted median filter [31], [32] and center median filters [33], [34]. However, these median filters still fail to preserve image edges when the IN density is high. Also, they cannot well handle the mixture of IN and AWGN. To better preserve the image edges and deal with mixed noise or impulse noise, variational methods [35]–[38] that recover the clean image by minimizing an energy functional, have been proposed. There are generally two types of variational methods for removing mixture noise. The first type methods are two-stages methods [10], [12], [39]–[42], which first detect the locations of the damaged pixels and then treat the denoising problem as an image inpainting problem. Obviously, the performance of this type methods relies on the accuracy of the detection of the damaged pixels. The second type methods treat the IN as outliers and don't explicitly detect the locations of the damaged pixels corrupted by the IN. Instead, robust fidelity terms based on ℓ_1 norm [43], [44] has been proposed to better adapt to the outliers. Coupled with an appropriate image regularizer (e.g., total variation, sparsity regularizers), these methods are effective in removing the IN. However, as the AWGN has not been considered in the fidelity term, the AWGN cannot be removed effectively by these methods. Such limitation has been addressed in the work of [13], where a reweighted ℓ_2 norm has been proposed for the fidelity term. By assigning smaller weights to pixels corrupted with IN and larger weights to pixels corrupted by AWGN, the method of [13] can simultaneously handle the IN and AWGN. However, an initial detection of the locations of the pixels corrupted by IN is required. Hence, its performance still depends on the accuracy of the initial detection of the damaged pixels to some extent. Considering the fact that the distributions of the IN are generally sparse,

the ℓ_p norm ($p = 0, 1$) has been used to model the IN [11], [14]. Coupled with an appropriate regularizer (e.g., TV and low-rank regularizer), both the clean image and the IN can be estimated via alternatively minimization, which consists of solving the image inpainting problem and the estimation of the IN by hard or soft thresholding (corresponding to ℓ_0 and ℓ_1 norm, respectively). These methods have shown promising denoising performance for removing mixed noise. However, the selection of appropriate thresholds such that the IN components can be accurately estimated from the image pixels remains a challenging task.

In this paper, we propose an effective mixed IN and AWGN removal method based on Laplacian scale mixture (LSM) modeling. In the proposed method, the IN is characterized with the LSM distribution, where both the hidden variables and the IN are jointly estimated from the observed noisy image. Compared with the previous ℓ_p ($p = 0, 1$) norm based methods that requires the selection of the threshold for determining the IN, the proposed method can adaptively learn the thresholds from the observed noisy image, leading to substantially improvements of the denoising performance. To exploit the nonlocal self-similarity of natural images, a nonlocal low-rank regularizer (NLR) is adopt to regularize the image denoising process. Experimental results on simulated noisy images show that the proposed method outperforms current state-of-the-art mixed noise removal methods.

The rest of the paper is organized as follows. The proposed denoising model with LSM modeling is presented in Sec. II, followed by the detailed optimization algorithm in Sec. III. Sec. IV presents the experimental results. Sec. V concludes the paper.

II. PROPOSED MIXED NOISE REMOVAL METHOD

A. Noise Model

We consider the mixture of the AWGN and IN. Two common types of IN are considered, i.e., the SPIN and RVIN. Assuming that the image pixel values are in the range of $[g_{min}, g_{max}]$, the pixels corrupted by SPIN can then only take two values g_{min} and g_{max} , while the pixels corrupted by RVIN can take any random value uniformly distributed in the range of $[g_{min}, g_{max}]$. Then, the observed noisy image can be modeled as

$$y_i = \begin{cases} x_i + w_i, & i \in \Omega \\ n_i, & i \in \Omega^C \end{cases} \quad (3)$$

where y_i , x_i and w_i denote the noisy pixel, the original pixel and the AWGN at position i , respectively, Ω and Ω^C denote the set of pixel positions where the pixels are corrupted by AWGN and IN, respectively, and n_i is the IN. In this paper, we consider three types of mixture noise: (1) AWGN + SPIN; (2) AWGN + RVIN; (3) AWGN + SPIN + RVIN. For all three cases, the observed noisy image $\mathbf{y} \in \mathbb{R}^N$ corrupted with mixed noise can be modeled as

$$\mathbf{y} = \mathbf{x} + \mathbf{s} + \mathbf{w}, \quad (4)$$

where $\mathbf{s} \in \mathbb{R}^N$ denotes the outlier component caused by IN. Comparing with Eq. (3), it is easy to show that $s_i = 0, i \in \Omega$

and $s_i = y_i - x_i - w_i, i \in \Omega^C$. Then, removing the mixed noise is formulated as a joint estimation problem of the outlier component $\mathbf{s}$ and the original image $\mathbf{x}$.

B. Laplacian scale mixture modeling of Impulse Noise

For the observation model of Eq. (4), we aim to jointly estimate the clean image $\mathbf{x}$, the outlier component $\mathbf{s}$ and the variance σ_w of the AWGN from the observed noisy image $\mathbf{y}$. This can be formulated as the following maximum a posterior (MAP) problem

$$\begin{aligned} (\mathbf{x}, \mathbf{s}, \sigma_w) &= \operatorname{argmax} \log P(\mathbf{y}|\mathbf{x}, \mathbf{s}, \sigma_w) P(\mathbf{x}, \mathbf{s}, \sigma_w) \\ &= \operatorname{argmax} \log P(\mathbf{y}|\mathbf{x}, \mathbf{s}, \sigma_w) + \log P(\mathbf{x}, \mathbf{s}) + \log P(\sigma_w) \\ &= \operatorname{argmax} \log P(\mathbf{y}|\mathbf{x}, \mathbf{s}, \sigma_w) + \log P(\mathbf{x}) + \log P(\mathbf{s}), \end{aligned} \quad (5)$$

where we have used the fact that the outlier components $\mathbf{s}$, the original image $\mathbf{x}$ and the noise variance σ_w are independent with each other, and set $P(\sigma_w) \propto 1$, which is an improper prior [45]. It means that σ_w has equal probability over the range $[0, \infty)$.

$P(\mathbf{y}|\mathbf{x}, \mathbf{s}, \sigma_w)$ is the likelihood term characterized by the Gaussian distribution with variance σ_w^2 . Note that the outlier $s_i = 0$ for the pixels corrupted with AWGN. For pixels corrupted with IN, $s_i = y_i - x_i - w_i$. Due to the pixel value range clipping, the corrupted pixel y_i is always in the range of $[g_{min}, g_{max}]$ no matter how the actual dynamic range of RVIN is. This turns out that the distribution of the outlier s_i will not be a multimodal or uniform distribution. Instead, the outlier s_i has the Laplacian-like distribution. In Fig. 1, we plot the empirical distributions of the outlier components $\mathbf{s}$ on the *Lena* image. The noisy images are generated by adding the mixed noise to the original *Lena* image. From Fig. 1, it can be seen that the empirical distributions of $\mathbf{s}$ can be characterized by Laplacian distributions, while the Gaussian distributions have much larger fitting errors. It is easy to verify that if $P(\mathbf{s})$ is chosen to be an identically independent distributed (i.i.d) zero-mean Laplacian distribution $P(s_i) = \frac{1}{2\theta} e^{-|s_i|/\theta}$, the MAP estimator of Eq. (5) leads to the following objective function

$$\begin{aligned} (\mathbf{x}, \mathbf{s}, \sigma_w) &= \operatorname{argmin}_{\mathbf{x}, \mathbf{s}, \sigma_w} \frac{1}{2\sigma_w^2} \|\mathbf{y} - \mathbf{x} - \mathbf{s}\|_2^2 + \lambda \|\mathbf{s}\|_1 + \eta J(\mathbf{x}), \\ &\quad + N \log \sigma_w, \end{aligned} \quad (6)$$

where $\lambda = \frac{1}{\theta}$, $P(\mathbf{x}) \propto \frac{1}{c} \exp(-\eta J(\mathbf{x}))$ is set as a Gibbs distribution, and θ is the standard deviation of s_i . The outlier components $\mathbf{s}$ can be easily solved by soft-thresholding for an initial estimate of $\mathbf{x}$. Such ℓ_1 norm based modeling for IN has shown promising denoising performance for mixed noise removal [11], [12]. Instead of using ℓ_1 as a convex approximation of ℓ_0 , the ℓ_0 norm has also been proposed to model the IN [14]. It has been shown that the ℓ_0 norm can achieve better results than ℓ_1 . By selecting an appropriate thresholds λ , the outlier component $\mathbf{s}$ can be accurately estimated. However, in both ℓ_1 and ℓ_0 based methods, it is often difficult to choose an appropriate regularization parameter λ , which depends on the noise level of IN.

In this paper, for each outlier component s_i , we introduce a hyperprior (i.e. the Jeffreys' prior [46]) on the scale parameter θ_i (i.e., the standard deviation) of the Laplacian distribution. Then, the distribution of $s_i = \alpha_i \theta_i$ is a Laplacian scale mixture (LSM), where α_i is a random variable that has Laplacian distribution with scale 1. The LSM is an analogy to the classic Gaussian scale mixture (GSM) [47], [48], which has been proven to be very effective for modeling sparse coefficients. Here, we propose to model the outliers with the LSM model. The advantage of the LSM modeling over the factorial Laplacian prior is that the scale parameters θ_i along with the coefficients α_i can be jointly estimated from the input noisy image. In contrast, the scale parameters θ_i in the factorial Laplacian prior have to be determined by hand. Assuming that each s_i is i.i.d and θ_i and α_i are independent, the LSM modeling of s_i can be formulated as

$$P(\mathbf{s}) = \prod_i P(s_i), \quad P(s_i) = \int_0^\infty P(s_i|\theta_i) P(\theta_i) d\theta_i. \quad (7)$$

Generally, there is no analytical expression of $P(s_i)$ for most of choices of $P(\theta_i)$ and thus is difficult to compute $\mathbf{s}$ with MAP. However, this difficulty can be avoided by using a joint prior model of $P(\mathbf{s}, \boldsymbol{\theta})$. By substituting $P(\mathbf{s}, \boldsymbol{\theta})$ into the MAP estimator of Eq. (5), we obtain

$$\begin{aligned} (\mathbf{x}, \mathbf{s}, \boldsymbol{\theta}, \sigma_w) &= \operatorname{argmax}_{\mathbf{x}, \mathbf{s}, \boldsymbol{\theta}, \sigma_w} \log P(\mathbf{y}|\mathbf{x}, \mathbf{s}, \sigma_w) + \log P(\mathbf{s}|\boldsymbol{\theta}) \\ &\quad + \log P(\boldsymbol{\theta}) + \log P(\mathbf{x}). \end{aligned} \quad (8)$$

For sparsity prior $P(\boldsymbol{\theta})$ of the hidden variable, we use the i.i.d. noninformative Jeffrey's prior $P(\theta_i) = \frac{1}{\theta_i}$ [46]. With the Jeffrey's prior, the MAP estimator of Eq. (8) can be expressed as

$$\begin{aligned} (\mathbf{x}, \mathbf{s}, \boldsymbol{\theta}, \sigma_w) &= \operatorname{argmin}_{\mathbf{x}, \mathbf{s}, \boldsymbol{\theta}, \sigma_w} \frac{1}{2\sigma_w^2} \|\mathbf{y} - \mathbf{x} - \mathbf{s}\|_2^2 + \sqrt{2} \sum_i \frac{|s_i|}{\theta_i} \\ &\quad + 2 \sum_i \log \theta_i + \eta J(\mathbf{x}) + N \log \sigma_w. \end{aligned} \quad (9)$$

In LSM, we have $\mathbf{s} = \boldsymbol{\Lambda} \boldsymbol{\theta}$, where $\boldsymbol{\Lambda} = \operatorname{diag}(\alpha_i) \in \mathbb{R}^{N \times N}$. Note that the Jeffrey's prior $P(\theta_i) = \frac{1}{\theta_i}$ is numerical unstable. For numerical stability, we introduce a small constant ϵ into $P(\theta_i)$, as $P(\theta_i) = \frac{1}{\theta_i + \epsilon}$. Then Eq. (9) can be rewritten as

$$\begin{aligned} (\mathbf{x}, \boldsymbol{\alpha}, \boldsymbol{\theta}, \sigma_w) &= \operatorname{argmin}_{\mathbf{x}, \boldsymbol{\alpha}, \boldsymbol{\theta}, \sigma_w} \frac{1}{2\sigma_w^2} \|\mathbf{y} - \mathbf{x} - \boldsymbol{\Lambda} \boldsymbol{\theta}\|_2^2 + \sqrt{2} \sum_i |\alpha_i| \\ &\quad + 2 \sum_i \log(\theta_i + \epsilon) + \eta J(\mathbf{x}) + N \log \sigma_w. \end{aligned} \quad (10)$$

With the LSM prior, the estimation of the outlier component $\mathbf{s}$ is then translated into the joint estimation of $\boldsymbol{\alpha}$ and $\boldsymbol{\theta}$.

C. Nonlocal low-rank regularization

In addition to the modeling of the outliers $\mathbf{s}$, the selection of the prior image model $P(\mathbf{x})$ is also of great importance. In recent years, the patch based nonlocal low-rank model has proven to be a very effective tool for image restoration [8], [10], [11], [49]. In this paper, we adopt the nonlocal

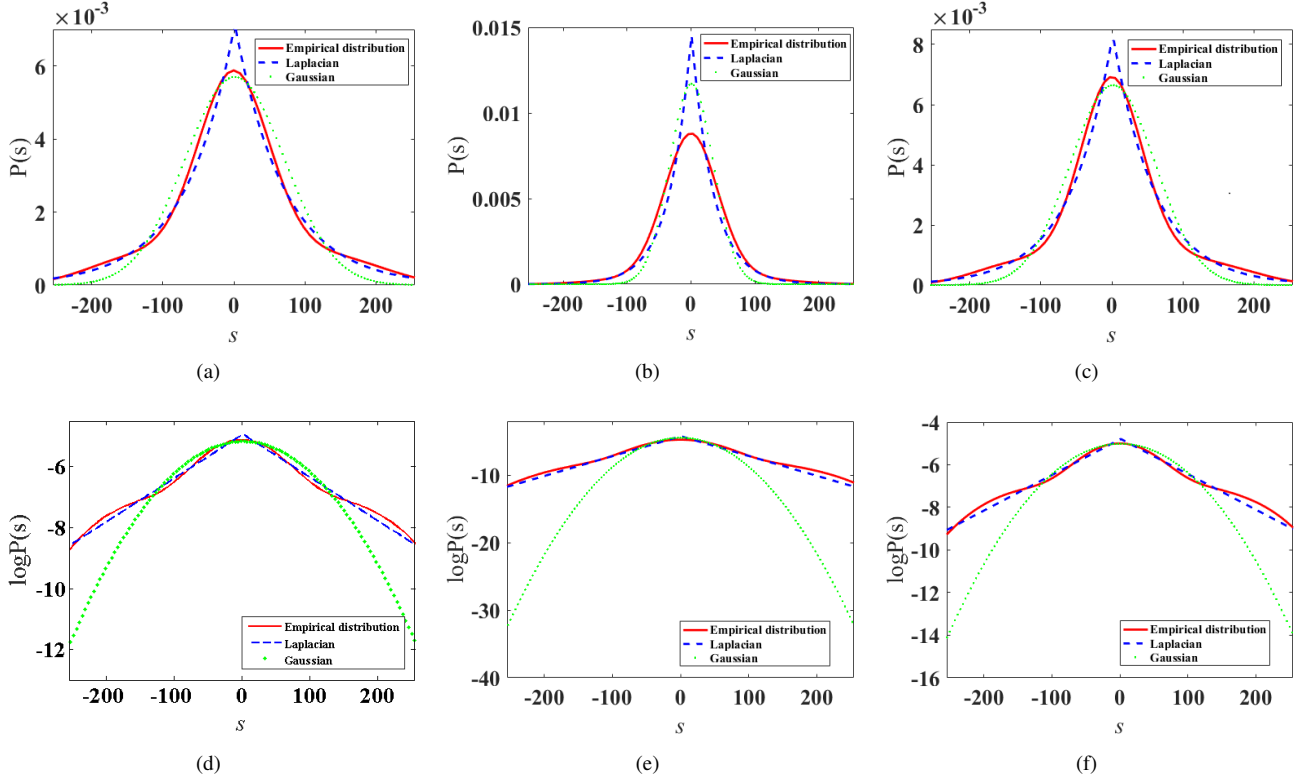


Fig. 1. Empirical distributions of s on simulated noisy *Lena* image. The noisy images are generated by adding the mixed noise to original *Lena* image: (a) AWGN + SPIN ($\sigma_w = 10$, $s = 30\%$); (b) AWGN + RVIN ($\sigma_w = 10$, $r = 15\%$); (c) AWGN + SPIN + RVIN ($\sigma_w = 10$, $s = 20\%$, $r = 15\%$). (d)-(f) show the same distributions of (a)-(c) in log domain, respectively.

low-rank regularization model proposed in our previous work [49], which can also be formulated as a MAP estimator. Let $\tilde{\mathbf{R}}_j \mathbf{x} \doteq [\mathbf{R}_{j_0} \mathbf{x}, \mathbf{R}_{j_1} \mathbf{x}, \dots, \mathbf{R}_{j_{m-1}} \mathbf{x}] \in \mathbb{R}^{n \times m}$ denote the matrix formed by the set of similar patches for exemplar image patch $\mathbf{x}_j \in \mathbb{R}^n$ of size $\sqrt{n} \times \sqrt{n}$ centered at position j , where $\tilde{\mathbf{R}}_j$ denote the operator extracting the similar patches. For each exemplar patch $\mathbf{x}_j$, the sets of similar patches can be formed by block matching, i.e., selecting the first m closest patches to $\mathbf{x}_j$ (including itself). Assuming that each group of similar patches are independent, the low-rank image prior can be expressed as

$$P(\mathbf{x}) \propto \prod_j P(\tilde{\mathbf{R}}_j \mathbf{x}), \quad P(\tilde{\mathbf{R}}_j \mathbf{x}) \propto \frac{1}{c} \exp(-\eta \text{rank}(\tilde{\mathbf{R}}_j \mathbf{x})). \quad (11)$$

By substituting Eq. (11) into Eq. (5) and combining with Eq. (10), we have

$$\begin{aligned} (\mathbf{x}, \boldsymbol{\alpha}, \boldsymbol{\theta}, \sigma_w) = \underset{\mathbf{x}, \boldsymbol{\alpha}, \boldsymbol{\theta}, \sigma_w}{\text{argmin}} \quad & \frac{1}{2\sigma_w^2} \|\mathbf{y} - \mathbf{x} - \boldsymbol{\Lambda} \boldsymbol{\theta}\|_2^2 + \sqrt{2} \sum_i |\alpha_i| \\ & + 2 \sum_i \log(\theta_i + \epsilon) + \eta \sum_j \text{rank}(\tilde{\mathbf{R}}_j \mathbf{x}) + N \log \sigma_w. \end{aligned} \quad (12)$$

As minimizing the $\text{rank}(\cdot)$ is a NP-hard problem, the nuclear norm $\|\cdot\|_*$ is often used as a convex surrogate of $\text{rank}(\cdot)$. Here, for better performance, a smooth but nonconvex surrogate of the rank is used. Specifically, the $\log \det(\cdot)$ function used in

[50]

$$\begin{aligned} L(\mathbf{X}, \epsilon) &= \log \det((\mathbf{X} \mathbf{X}^T)^{1/2} + \epsilon \mathbf{I}) \\ &= \log \det(\boldsymbol{\Sigma}^{1/2} + \epsilon \mathbf{I}) \\ &= \sum_{r=1}^{r_0} \log(\sigma_r(\mathbf{X}) + \epsilon) \end{aligned} \quad (13)$$

is used as a nonconvex surrogate of rank, where $\mathbf{X}$ is a matrix of size $\mathbb{R}^{n \times m}$, $\boldsymbol{\Sigma}$ is the diagonal matrix whose diagonal elements are eigenvalues of matrix $\mathbf{X} \mathbf{X}^T$, i.e. $\mathbf{X} \mathbf{X}^T = \mathbf{U} \boldsymbol{\Sigma} \mathbf{U}^{-1}$, $\boldsymbol{\Sigma}^{1/2}$ is the diagonal matrix whose diagonal elements are the singular values of the matrix $\mathbf{X}$, and $\sigma_r(\mathbf{X})$ denotes the r^{th} singular value of $\mathbf{X}$ and $r_0 = \min(n, m)$. It has shown in [49] that the $\log \det(\cdot)$ has better performance than nuclear norm for low-rank approximation. Then, Eq. (12) can be rewritten as

$$\begin{aligned} (\mathbf{x}, \boldsymbol{\alpha}, \boldsymbol{\theta}, \sigma_w) = \underset{\mathbf{x}, \boldsymbol{\alpha}, \boldsymbol{\theta}, \sigma_w}{\text{argmin}} \quad & \frac{1}{2\sigma_w^2} \|\mathbf{y} - \mathbf{x} - \boldsymbol{\Lambda} \boldsymbol{\theta}\|_2^2 + \sqrt{2} \sum_i |\alpha_i| \\ & + 2 \sum_i \log(\theta_i + \epsilon) + \eta \sum_j L(\tilde{\mathbf{R}}_j \mathbf{x}, \epsilon) + N \log \sigma_w. \end{aligned} \quad (14)$$

In the proposed objective function of Eq. (14), the outlier components s can be adaptively estimated from the observed noisy image, without the needs of manually selection of the regularization parameters. With the nonlocal low-rank regularization, the group sparsity of similar patches can be effectively exploited. Thus, better denoising performance can be achieved.

In the next section, we will show that the proposed objective function can be solved by alternative optimization.

III. OPTIMIZATION ALGORITHM

A straightforward approach for solving the objective function is to use the alternatively optimization. For an initial estimate of $\mathbf{x}$, we solve for the outliers by alternatively optimizing $\boldsymbol{\theta}$ and $\boldsymbol{\alpha}$, and update $\mathbf{x}$ with fixed $\mathbf{s} = \boldsymbol{\Lambda}\boldsymbol{\theta}$. The noise variance σ_w of AWGN is also iteratively estimated based on the updated estimates of $\mathbf{x}$ and $\mathbf{s}$.

A. Solving the $\boldsymbol{\theta}$ -subproblem

For fixed $\mathbf{x}$, $\boldsymbol{\alpha}$ and σ_w , the positive hidden variables $\boldsymbol{\theta}$ can be solved by minimizing

$$\boldsymbol{\theta} = \underset{\boldsymbol{\theta}}{\operatorname{argmin}} \|\mathbf{y} - \mathbf{x} - \boldsymbol{\Lambda}\boldsymbol{\theta}\|_2^2 + 4\sigma_w^2 \sum_i \log(\theta_i + \epsilon), \quad s. t. \theta_i \geq 0. \quad (15)$$

Equivalently, Eq. (15) can be rewritten as

$$\begin{aligned} \boldsymbol{\theta} = \underset{\boldsymbol{\theta}}{\operatorname{argmin}} \sum_i (y_i - x_i - \alpha_i \theta_i)^2 + 4\sigma_w^2 \sum_i \log(\theta_i + \epsilon) \\ s. t. \theta_i \geq 0, \end{aligned} \quad (16)$$

Therefore, each θ_i can be independently solved by solving a scalar minimization problem, as

$$\theta_i = \underset{\theta_i}{\operatorname{argmin}} a_i \theta_i^2 + b_i \theta_i + c \log(\theta_i + \epsilon), \quad s. t. \theta_i \geq 0, \quad (17)$$

where $a_i = \alpha_i^2$, $b_i = 2\alpha_i(x_i - y_i)$ and $c = 4\sigma_w^2$. Though Eq. (17) is nonconvex, the closed-form solution can be derived by taking $\frac{f(\theta_i)}{d\theta_i} = 0$, where $f(\theta_i)$ denotes the right hand side of Eq. (17). It is easy to show that the solution to Eq. (17) is given by

$$\theta_i = \begin{cases} 0, & \text{if } (2a_i\epsilon + b_i)^2 / (16a_i^2) - (b_i\epsilon + c) / (2a_i) < 0 \\ t_i, & \text{otherwise,} \end{cases} \quad (18)$$

where $t_i = \underset{\theta_i}{\operatorname{argmin}} \{f(0), f(\theta_{i,1}), f(\theta_{i,2})\}$, where $\theta_{i,1}$ and $\theta_{i,2}$ are two stationary points of $f(\theta_i)$, i.e.,

$$\begin{aligned} \theta_{i,1} &= -\frac{2a_i\epsilon + b_i}{4a_i} + \sqrt{\frac{(2a_i\epsilon + b_i)^2}{16a_i^2} - \frac{b_i\epsilon + c}{2a_i}} \\ \theta_{i,2} &= -\frac{2a_i\epsilon + b_i}{4a_i} - \sqrt{\frac{(2a_i\epsilon + b_i)^2}{16a_i^2} - \frac{b_i\epsilon + c}{2a_i}}. \end{aligned} \quad (19)$$

B. Solving the $\boldsymbol{\alpha}$ -subproblem

For fixed $\boldsymbol{\theta}$, the Laplaican coefficients can be updated via solving

$$\boldsymbol{\alpha} = \underset{\boldsymbol{\alpha}}{\operatorname{argmin}} \sum_i (y_i - x_i - \theta_i \alpha_i)^2 + 2\sqrt{2}\sigma_w^2 \sum_i |\alpha_i|. \quad (20)$$

Then each α_i can be solved by optimizing a scalar minimization problem, i.e.,

$$\alpha_i = \underset{\alpha_i}{\operatorname{argmin}} (y_i - x_i - \theta_i \alpha_i)^2 + 2\sqrt{2}\sigma_w^2 |\alpha_i|, \quad (21)$$

which admits a closed-form solution

$$\alpha_i = \mathcal{S}_{\tau_i}((y_i - x_i)/(\theta_i + \epsilon)), \quad (22)$$

where $\mathcal{S}_{\tau_i}(\cdot)$ denotes a soft-thresholding operator with threshold $\tau_i = \frac{2\sqrt{2}\sigma_w^2}{(\theta_i + \epsilon)^2}$ and ϵ is used for numerical stability.

C. Solving the $\mathbf{x}$ -subproblem

With the updated outlier components $\mathbf{s} = \boldsymbol{\Lambda}\boldsymbol{\theta}$, the image $\mathbf{x}$ can be estimated by solving

$$\mathbf{x} = \underset{\mathbf{x}}{\operatorname{argmin}} \frac{1}{2\sigma_w^2} \|\mathbf{y} - \mathbf{s} - \mathbf{x}\|_2^2 + \eta \sum_j L(\tilde{\mathbf{R}}_j \mathbf{x}, \epsilon). \quad (23)$$

To facilitate the optimization, we introduce an auxiliary variable $\mathbf{L}_j$, i.e.,

$$\begin{aligned} (\mathbf{x}, \mathbf{L}_j) = \underset{\mathbf{x}, \mathbf{L}_j}{\operatorname{argmin}} \frac{1}{2\sigma_w^2} \|\mathbf{y} - \mathbf{s} - \mathbf{x}\|_2^2 + \eta \sum_j L(\mathbf{L}_j, \epsilon), \\ s. t. \mathbf{L}_j = \tilde{\mathbf{R}}_j \mathbf{x}. \end{aligned} \quad (24)$$

Using the alternative direction multiplier method (ADMM) [51]–[53], we can convert the constrained optimization problem into an unconstrained one, i.e.,

$$\begin{aligned} (\mathbf{x}, \mathbf{L}_j) = \underset{\mathbf{x}, \mathbf{L}_j}{\operatorname{argmin}} \frac{1}{2\sigma_w^2} \|\mathbf{y} - \mathbf{s} - \mathbf{x}\|_2^2 \\ + \mu \sum_j \|\tilde{\mathbf{R}}_j \mathbf{x} - \mathbf{L}_j + \frac{\mathbf{U}_j}{2\mu}\|_F^2 + \eta \sum_j L(\mathbf{L}_j, \epsilon), \end{aligned} \quad (25)$$

where $\mathbf{U}_j$ is the augmented Lagrangian multiplier. Then, solving Eq. (25) consists of the following alternative iterations:

$$\begin{aligned} \mathbf{x} = \underset{\mathbf{x}}{\operatorname{argmin}} \frac{1}{2\sigma_w^2} \|\mathbf{y} - \mathbf{s} - \mathbf{x}\|_2^2 + \mu \sum_j \|\tilde{\mathbf{R}}_j \mathbf{x} - \mathbf{L}_j + \frac{\mathbf{U}_j}{2\mu}\|_F^2, \\ \mathbf{L}_j = \underset{\mathbf{L}_j}{\operatorname{argmin}} \mu \|\tilde{\mathbf{R}}_j \mathbf{x} - \mathbf{L}_j + \frac{\mathbf{U}_j}{2\mu}\|_F^2 + \eta \sum_j L(\mathbf{L}_j, \epsilon). \end{aligned} \quad (26)$$

Using Eq. (13), the $\mathbf{L}_j$ -subproblem in Eq.(26) can be further rewritten as

$$\mathbf{L}_j = \underset{\mathbf{L}_j}{\operatorname{argmin}} \mu \|\tilde{\mathbf{R}}_j \mathbf{x} - \mathbf{L}_j + \frac{\mathbf{U}_j}{2\mu}\|_F^2 + \eta \sum_{r=1}^{r_0} \log(\sigma_r(\mathbf{L}_j) + \epsilon), \quad (27)$$

where $\sigma_r(\mathbf{L}_j)$ denotes the r^{th} singular value of $\mathbf{L}_j$ and $r_0 = \min(n, m)$. The $\mathbf{x}$ -subproblem in Eq. (26) can be easily solved in a closed-form solution, i.e.,

$$\mathbf{x} = (2\sigma_w^2 \mu \sum_j \tilde{\mathbf{R}}_j^\top \tilde{\mathbf{R}}_j + \mathbf{I})^{-1} (\mathbf{y} - \mathbf{s} + 2\sigma_w^2 \mu \sum_j \tilde{\mathbf{R}}_j^\top (\mathbf{L}_j - \frac{\mathbf{U}_j}{2\mu})), \quad (28)$$

where $\tilde{\mathbf{R}}_j^\top \mathbf{L}_j \doteq \sum_{r=0}^{m-1} \mathbf{R}_{j_r}^\top \mathbf{x}_{j_r}$ and $\tilde{\mathbf{R}}_j^\top \tilde{\mathbf{R}}_j \doteq \sum_{r=0}^{m-1} \mathbf{R}_{j_r}^\top \mathbf{R}_{j_r}$, where $\mathbf{R}_{j_r}$ denotes the matrix extracting the image patch at position j_r . The matrix to be inverted in Eq. (28) is a diagonal matrix. Hence, the $\mathbf{x}$ solution of Eq. (28) can be computed efficiently. As shown in [49], the

logdet based matrix low-rank approximation problem can be iteratively solved, i.e.,

$$\mathbf{L}_j^{(k+1)} = \mathbf{U}(\tilde{\Sigma} - \frac{\eta}{2\mu} \text{diag}(\mathbf{w}^{(k)}))_+ \mathbf{V}^\top, \quad (29)$$

where $\mathbf{U}\tilde{\Sigma}\mathbf{V}^\top$ is the SVD of $\tilde{\mathbf{R}}_j\mathbf{x}$, and $w_r^{(k)} = \frac{1}{(\sigma_r^{(k)} + \epsilon)}$, wherein $\sigma_r^{(k)}$ is the estimated r^{th} singular value obtained in the k^{th} iteration. The augmented Lagrangian multiplier $\mathbf{U}$ is updated via

$$\mathbf{U}_j^{(k+1)} = \mathbf{U}_j^{(k)} + \mu(\tilde{\mathbf{R}}_j\mathbf{x} - \mathbf{L}_j). \quad (30)$$

Note that since the logdet function in Eq. (24) is nonconvex, the convergence of ADMM algorithm on nonconvex objective functions is not guaranteed in theory. However, recently, it has been proven in [54] that the ADMM algorithm converges on some classes of nonconvex function, e.g., the Lipschitz differentiable functions. It is easy to verify that the logdet function is also Lipschitz differentiable and thus the ADMM algorithm for solving Eq. (25) will converge. This is consistent with our empirical observation, as shown in Fig. 2(a). In Fig. 2(a) we show that the curve of the objective function value of Eq.(25) as a function of iteration number on test *Lena* image corrupted with AWGN+SPIN ($\sigma_w = 20$, $s = 40\%$). It can be seen that the ADMM algorithm converges generally in only 5-7 iterations.

D. Solving the σ_w -subproblem

With updated $\mathbf{x}$ and $\mathbf{s}$, the noise variance σ_w of AWGN can be estimated by solving

$$\sigma_w = \underset{\sigma_w}{\text{argmin}} \frac{1}{2\sigma_w^2} \|\mathbf{y} - \mathbf{x} - \mathbf{s}\|_2^2 + N \log \sigma_w. \quad (31)$$

By taking the $\frac{g(\sigma_w)}{d\sigma_w} = 0$, where $g(\sigma_w)$ denotes the right hand side of Eq. (31), closed-form solution can be obtained as

$$\sigma_w = \sqrt{\|\mathbf{y} - \mathbf{x} - \mathbf{s}\|_2^2 / N}. \quad (32)$$

The proposed mixed noise removal method based on LSM modeling and nonlocal low-rank regularization is summarized in **Algorithm 1**. In **Algorithm 1**, we apply the median filtering to the noisy image for an initial estimate of $\mathbf{x}$, which is used for similar patch grouping. The patch grouping is updated in every T_0 iteration to save computational complexity. The inner loop for low-rank approximation converges fast. Fig.2(b) shows the curve of the objective function value of Eq.(14) as a function of iteration number t on test image *Lena* corrupted with AWGN+SPIN ($\sigma_w = 20$, $s = 40\%$). It can be seen that **Algorithm 1** converges fast(usually after 4 iterations).

IV. EXPERIMENTAL RESULTS

In this section, we present the experimental results of the propose mixed noise removal method based on Laplacian scale modeling (LSM) and nonlocal low-rank regularization (denoted as LSM-NLR). The basic parameters of the proposed method is set as follows: the patch size is set as 8×8 ; the number of similar patches $m = 40$; exemplar image patches are extracted in every 4 pixels along both horizontal and

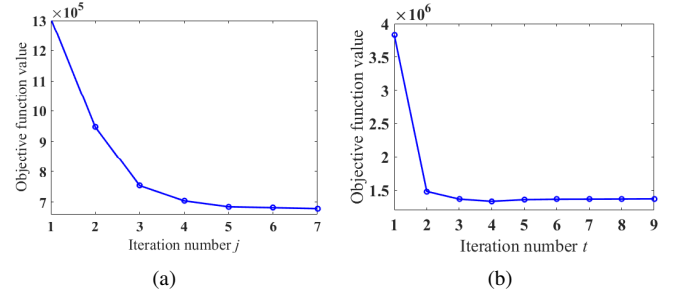


Fig. 2. The curve of the objective function of Eq.(23) and (14). (a) The curve of the objective function of Eq.(23) versus iteration number j ; (b) The curve of the objective function of Eq.(14) versus iteration number t .

Algorithm 1 Mixed noise removal via LSM modeling and NLR

• Initialization:

- (1) Set parameter η and $\mathbf{s} = \mathbf{0}$;
- (2) Initialize $\mathbf{x}^{(0)}$ via median filtering;
- (3) Grouping similar patches for each $\mathbf{x}_j$ using $\mathbf{x}^{(0)}$;

• Outer loop: for $t = 1, 2, \dots, T$ do

- (1) Update σ_w via Eq. (32)
- (2) Update θ via Eq. (18);
- (3) Update α via Eq. (22);
- (4) **Inner loop** (Low-rank approximation, Solving Eq. (25)): for $j = 1, 2, \dots, J$ do

Inner loop for $k = 1, 2, \dots, K$ do

- (a) Update the weights $w_r = 1/(\sigma_r(\mathbf{L}_j^{(k-1)}) + \epsilon)$;
- (b) Singular value thresholding via Eq. (29);
- (c) Output $\mathbf{L}_j = \mathbf{L}_j^{(K)}$ if $k = K$.

End for

- (5) Update $\mathbf{x}$ via Eq. (28);
- (III) Update $\mathbf{U}^{(j+1)}$ via Eq. (30)
- (IV) Update $\mu := \rho\mu$ ($\rho > 1$);

End for

- (6) If $\text{mod}(t, T_0) = 0$, update the patch grouping;
- (7) Output the denoised image $\hat{\mathbf{x}} = \mathbf{x}^{(t)}$ if $t = T$.

End for

vertical directions; the small constant ϵ is set as $\epsilon = 10^{-5}$; the regularization parameter η is denoted as $c\sigma_w^2$ and $c = 6, 5, 7$ when $\sigma_w \leq 10$, $c = 6, 3, 7$ when $\sigma_w > 10$ for the mixture noise of AWGN+SPIN, AGWN+RVIN, AWGN+SPIN+RVIN respectively. *The source code of the proposed method can be downloaded from the project website*¹. We have compared the proposed method with several leading mixed noise removal methods, including the Two-phase deblurring/denoising (TPD) method [9], the sparse and low-rank regularization denoising (SLR) method [11], the weighted encoding with sparse nonlocal regularization (WESNR) method [13], and the well-known BM3D [3] method coupled with median filter (denoted as BM3D*)². To deal with the mixture noise, in the BM3D* method the adaptive median filter (AMF) [32] and adaptive center-weighted median filter (ACWMF) [34] are applied sequentially to remove the SPIN and RVIN, respectively,

¹http://see.xidian.edu.cn/faculty/wsdong/LSM_denoising_project.htm

²We thank the authors of [3], [9], [11], [13] for sharing their source codes.

followed by the BM3D method for removing the AWGN. To demonstrate the effectiveness of the proposed LSM, we also compared to the following variants of the proposed denoising method based on ℓ_p -norm ($p = 0, 1$) and nonlocal low-rank regularization (denoted as ℓ_0 -NLR and ℓ_1 -NLR, respectively), i.e.,

$$(\mathbf{x}, \mathbf{s}, \sigma_w) = \underset{\mathbf{x}, \mathbf{s}, \theta, \sigma_w}{\operatorname{argmin}} \frac{1}{2\sigma_w^2} \|\mathbf{y} - \mathbf{x} - \mathbf{s}\|_2^2 + \lambda \|\mathbf{s}\|_p + \eta \sum_j L(\tilde{\mathbf{R}}_j \mathbf{x}, \epsilon) + N \log \sigma_w, \quad (33)$$

which can also be solved by alternative optimization. The $\mathbf{s}$ component can be solved in closed-form solutions, i.e., $\mathbf{s} = \mathcal{H}_\tau(\mathbf{y} - \mathbf{x})$ and $\mathbf{s} = \mathcal{S}_\tau(\mathbf{y} - \mathbf{x})$ for the cases of $p = 0$ and $p = 1$, respectively, where $\mathcal{H}_\tau(\cdot)$ denotes the hard-thresholding with threshold $\tau = 2\lambda\sigma_w$. For fixed $\mathbf{s}$, $\mathbf{x}$ and σ_w can also be estimated using the optimization techniques presented in Sec. III (C) and (D).

A. Experiments on mixed noise of AWGN and IN

As shown in Fig. 3, 12 grayscale images of size 512×512 , including 8 classic test images widely used in the literature of image denoising and other four high-quality test images, are employed to simulate the noisy images. Three types of mixture noise, i.e., AWGN+SPIN, AWGN+RVIN, and AWGN+SPIN+RVIN are considered. For each experiment, we run all the test methods 20 times independently and report the mean results of the 20 outputs. The full-reference image quality assessment (IQA) methods including the PSNR and the perceptual quality metric FSIM [55] are used to evaluate the qualities of the denoised images. We also have used a blind IQA method to assess the denoising results since the blind IQA method focuses on the the perceptual quality of the human visual system (HVS). Due to its good performance, the Codebook Representation for No-Reference Image Assessment (CORNIA) ³ [56] is adopted.

To generate different noise levels, we vary the standard deviations of AWGN σ_w from 10 to 50, the SPIN ratio s with 10%, 20%, 30%, 40% and 50%, and the RVIN ratio with $r = 10\%, 20\%, 30\%, 40\%$ and 50%. Tables I and II show the average PSNR, FSIM and CORNIA results of the denoised images by the test methods on the set of 12 test images for mixture noise of AWGN+SPIN and AWGN+RVIN, respectively. From Tables I and II, we can see that the variants of the proposed methods, i.e., ℓ_1 -NLR and ℓ_0 -NLR methods outperform the TPD method [9], the SLR method [11], and the recent developed WESNR method [13]. This verifies the effectiveness of the sparse and low-rank modeling for mixture noise removal. It can also be seen that the ℓ_0 -NLR method performs better than the ℓ_1 -NLR method for the mixture noise of AWGN+RVIN. Such observation is consistent with those obtained in [14]. Obviously, the proposed LSM-NLR method consistently outperforms the ℓ_1 -NLR, ℓ_0 -NLR methods and other competing methods. The average PSNR gains over ℓ_1 -NLR and ℓ_0 -NLR methods can be up to 1.83 dB and 1.63

dB, respectively, demonstrating the advantages of the proposed LSM technique for SPIN and RVIN modeling over the conventional ℓ_1 and ℓ_0 based methods. In Figs. 4 and 5, we show parts of the denoised images by the test methods on the two types of mixed noise. It can be seen that the proposed method can reproduce much sharper image edges and preserve finer details than other competing methods.

For the mixture noise of AWGN+SPIN+RVIN, we didn't compared with the TPD method [9], as it cannot deal with the AWGN+SPIN+RVIN. Here we vary the standard deviations of AWGN σ_w with 5, 10, 15 and 20, the SPIN ratio s with 10%, 30%, 40% and 50%, and the RVIN ratio r with 5%, 10%, 15% and 30%. Table III shows the PSNR, FSIM and CORNIA results of the test methods. We can see that the proposed method still consistently outperforms other competing methods. Parts of the denoised images by the methods are shown in Fig. 6, from which it can be seen that the proposed method reproduces the most visually pleasant images with sharper edges and finer image details.

We have also conducted experiments on a standard datasets, i.e., the commonly used Berkeley Segmentation Dataset (BSD100) ⁴ containing 100 color images. For simplicity, we conducted experiments only on the luminance components of the test images of BSD100 dataset. We didn't include TPD method in the comparison study on BSD100 dataset as it can only deal with images with the same numbers of rows and columns. Tables IV, V and VI shows the average PSNR, FSIM and CORNIA results of the denoised images by the test methods on BSD100 dataset for mixture noise of AWGN+SPIN, AWGN+RVIN, AWGN+SPIN+RVIN respectively. One can see that the proposed method still consistently outperforms other competing methods. We have also conducted experiments on some of color images from BSD100 dataset. Obviously, better performance would be achieved if the color channels are denoised jointly. However, for the convenience of comparison study, in this experiment each color channel is denoised separately by different test methods. Fig. 7 shows parts of the denoised color images by the test methods on the case of AWGN+SPIN+RVIN. It can be seen that the proposed method outperforms other competing methods. Please refer to our project website for more experimental results.

B. Experiments for AWGN denoising

To verify the effectiveness of the low-rank regularizer, we also conducted experiments for AWGN denoising. For AWGN only, the proposed method reduces to the logdet-based low-rank approximation for AWGN denoising, while the WESNR method [13] reduces to the NCSR method [6]. The SLR method [11] is also a low-rank method using nuclear norm. The two-stage TPD method [9] is specifically designed for removing mixed noise and not suitable for removing Gaussian noise. Table VII shows the average denoising performances of the competing methods on the set of test images shown in Fig. 3. It can be seen that the performance of the proposed LSM-NLR is significant better than SLR. This is mainly due to the fact that it is quite difficult to select an appropriate threshold

³<http://www.umiacs.umd.edu/~pengye/research.html>

⁴<https://www2.eecs.berkeley.edu/Research/Projects/CS/vision/bsds/>

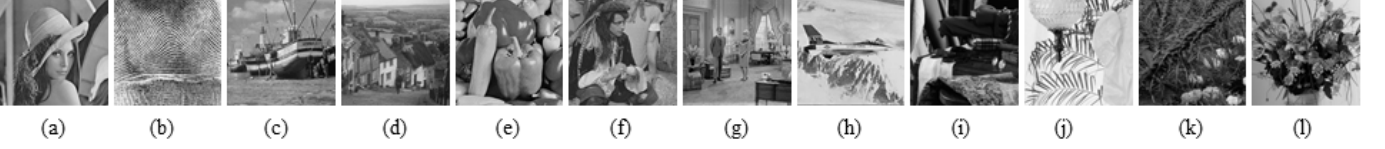


Fig. 3. The eight classic test images (a-h) and other four high-quality test images (i-l). From left to right and top to bottom: Lena, FG, Boat, Hill, Peppers, Man, Couple, AP, Cloth, Vase, Bush, Flower.

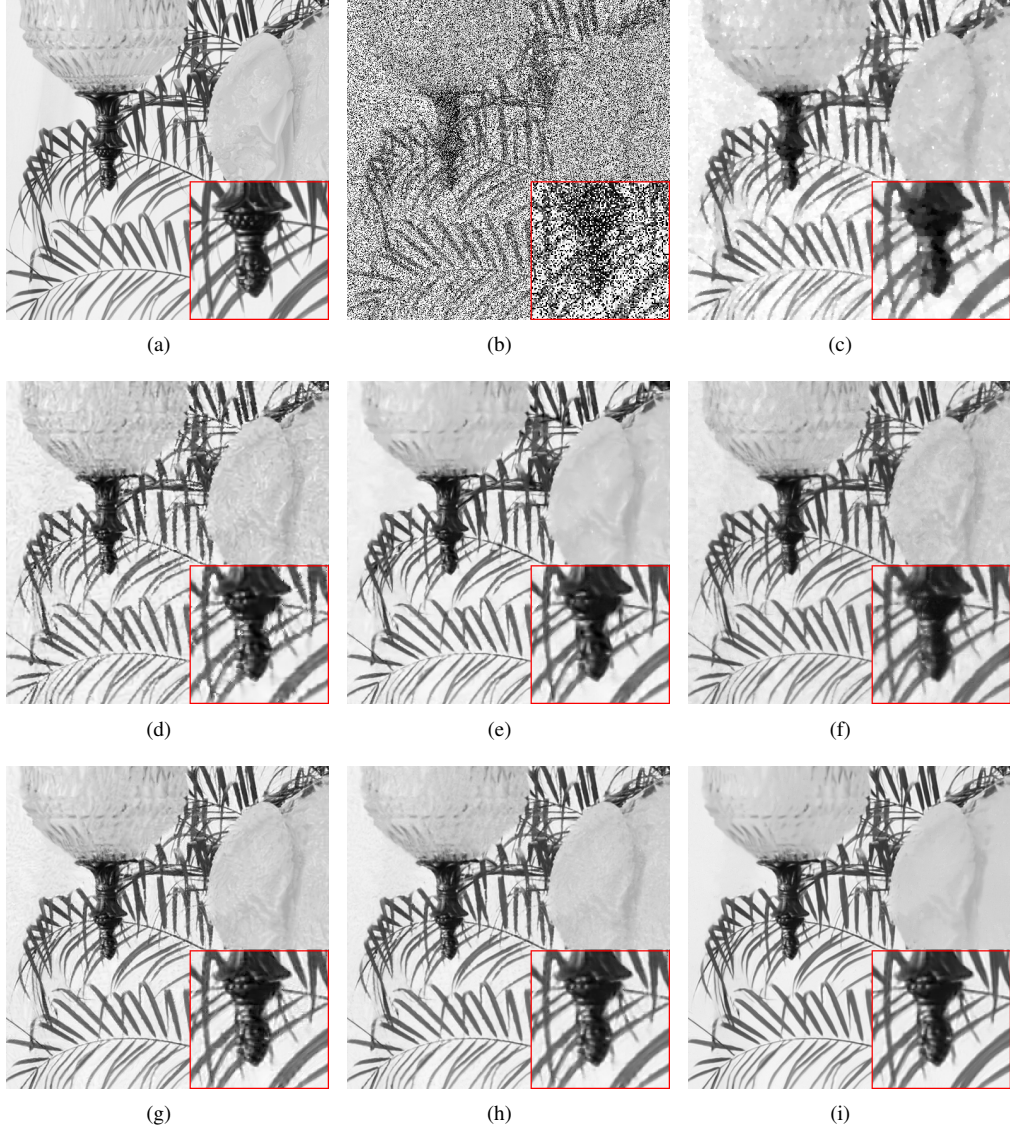


Fig. 4. Denoising results of different methods on test image *Vase* corrupted with AWGN+SPIN($\sigma_w = 20, s = 50\%$). (a) Original image; (b) Noisy image; images denoised by (c) TPD method [9] (PSNR= 23.58dB, FSIM= 0.8881, CORNIA= 38.88); (d) BM3D* [3] (PSNR= 24.50dB, FSIM= 0.9089, CORNIA= 35.61); (e) WESNR [13] (PSNR= 24.43dB, FSIM= 0.9235, CORNIA= 20.38); (f) SLR [11] (PSNR= 26.60dB, FSIM= 0.9396, CORNIA= 21.10) (g) ℓ_1 -NLR (PSNR= 27.56dB, FSIM= 0.9442, CORNIA= 22.52); (h) ℓ_0 -NLR (PSNR= 27.72dB, FSIM= 0.9464, CORNIA= 21.07); (i) proposed **LSM-NLR** (PSNR= **29.24dB**, FSIM= **0.9556**, CORNIA= **6.70**).

by hand for nuclear-norm based low-rank approximation. The LSM-NLR method is slightly better than the NCSR method, and is comparable with the well-known BM3D method.

C. Parameters selection

There are only three free parameters in the proposed method that need to be tuned, i.e., the patch size $\sqrt{n} \times \sqrt{n}$, number of

similar patches m , and the regularization parameter $\eta = \gamma\sigma_w^2$. Fig.8 shows the average PSNR curves as functions of the patch size n , the number of similar patches m , and the regularization parameter γ on the set test images shown in Fig.3 corrupted with AWGN+SPIN ($\sigma_w = 20, s = 40\%$), respectively. Similar observations can be observed for other experimental settings. From Fig.8(a) it can be seen that the performance of the proposed method is insensitive to the patch

TABLE I
THE AVERAGE PSNR(dB), FSIM (%) AND CORNIA RESULTS OF MIXED NOISE REMOVAL (AWGN+SPIN) ON THE SET OF 12 TEST IMAGES.

Method		TPD [9]	BM3D* [3]	WESNR [13]	SLR [11]	ℓ_1 -NLR	ℓ_0 -NLR	LSM-NLR
$\sigma_w = 10$	$s = 10\%$	30.60	31.81	31.38	32.20	32.50	32.84	33.60
		96.83	97.59	97.24	97.85	97.89	98.01	98.25
		39.03	27.91	27.68	31.82	29.22	19.79	19.03
	$s = 20\%$	29.82	30.83	31.18	31.52	31.95	32.18	32.80
		95.30	97.17	97.02	97.55	97.94	97.94	98.02
		39.95	30.96	29.41	29.26	28.63	18.45	20.10
	$s = 30\%$	28.92	29.65	30.66	30.73	31.50	31.87	32.37
		95.30	96.51	96.69	97.13	97.50	97.39	97.72
		41.35	36.00	31.86	29.66	28.78	20.18	17.61
	$s = 40\%$	27.95	28.32	29.91	29.77	30.62	30.79	31.60
		94.04	95.59	96.23	96.55	97.01	97.10	97.28
		43.56	39.42	32.28	29.52	28.17	22.32	17.25
	$s = 50\%$	26.80	26.93	28.95	28.65	29.75	29.89	30.60
		92.09	94.29	95.29	95.49	96.49	96.39	96.63
		46.75	46.22	35.10	30.01	29.18	26.60	17.80
$\sigma_w = 20$	$s = 10\%$	27.81	29.22	28.86	28.45	28.70	29.17	30.52
		93.48	95.01	93.97	94.97	95.76	95.85	95.86
		54.00	32.69	31.99	40.94	37.59	35.37	22.78
	$s = 20\%$	27.43	28.69	28.59	28.10	28.49	28.98	30.16
		92.91	94.40	93.73	94.62	95.40	95.50	95.59
		48.30	34.91	34.08	39.53	37.94	34.89	23.12
	$s = 30\%$	26.92	27.94	28.16	27.63	28.72	29.10	29.71
		92.08	93.49	93.39	94.15	94.92	94.98	95.05
		48.60	38.79	36.21	38.23	37.50	32.58	22.91
	$s = 40\%$	26.26	27.02	27.56	27.10	28.40	28.53	29.15
		90.82	92.21	92.84	93.52	94.20	94.14	94.72
		51.29	42.48	37.87	35.92	33.86	31.65	25.39
	$s = 50\%$	25.45	25.94	26.73	26.43	27.50	27.75	28.51
		88.91	90.41	91.99	92.61	93.32	93.38	93.76
		51.7	47.86	39.64	35.60	33.48	31.84	24.16
$\sigma_w = 30$	$s = 10\%$	25.87	27.51	27.22	25.74	26.72	27.21	28.55
		90.06	92.54	91.91	91.89	93.49	93.60	93.62
		52.14	36.87	37.37	50.52	43.52	41.12	27.72
	$s = 20\%$	25.61	27.03	26.80	25.46	26.59	27.09	28.31
		89.47	91.68	91.55	91.57	93.02	93.12	93.21
		52.46	38.69	40.68	47.44	40.50	38.73	27.16
	$s = 30\%$	25.25	26.34	26.18	25.07	26.45	26.91	27.82
		88.74	90.39	90.94	91.06	92.32	92.40	93.03
		55.26	41.65	42.16	45.22	39.06	35.88	32.20
	$s = 40\%$	24.70	25.51	25.48	24.60	26.22	26.59	27.40
		87.55	88.67	90.12	90.30	91.32	91.35	92.23
		55.90	45.46	43.61	43.57	36.72	35.67	29.20
	$s = 50\%$	23.96	24.51	24.52	24.04	25.85	26.13	26.70
		85.79	86.48	88.80	89.34	90.02	90.04	91.06
		56.65	50.41	44.95	39.98	36.65	34.60	31.30
$\sigma_w = 50$	$s = 10\%$	23.05	25.05	20.80	21.63	24.44	24.83	26.00
		83.52	88.45	82.29	86.65	89.31	89.36	90.04
		57.06	44.06	58.70	58.06	48.41	46.68	42.05
	$s = 20\%$	22.79	24.59	18.53	21.25	24.27	24.70	25.90
		83.15	87.25	77.61	86.08	88.51	88.52	89.61
		57.81	44.99	58.21	55.99	45.79	42.86	39.73
	$s = 30\%$	22.35	24.06	16.80	20.87	24.05	24.44	25.61
		82.57	85.69	73.38	85.48	87.37	87.28	88.56
		58.54	47.08	59.33	54.22	44.73	43.69	36.95
	$s = 40\%$	21.68	23.56	15.52	20.42	23.76	24.07	25.05
		81.68	83.46	69.71	84.65	85.79	85.62	87.15
		58.95	48.57	59.08	51.36	42.83	41.19	38.92
	$s = 50\%$	20.71	22.38	14.43	19.90	23.35	23.56	24.36
		80.18	80.68	66.08	83.64	83.69	83.50	85.44
		58.85	51.24	58.00	46.86	42.15	39.31	41.86

size. As patch size of 64 (i.e. 8×8) generally leads to the best performance, the patch size of 64 (i.e. 8×8) is adopted in our implementation. From Fig.8(b) we can see that the performance of the proposed method is also insensitive to the number of the similar patches m . For the tradeoff between the performance and the computational complexity, we set $m = 40$ in our implementation. From Fig.8(c), one can see that the performance of the proposed method is also robust to the varying of the values of η . In general, $\gamma = 5 \sim 6$ leads to the best performance, we set $\eta = 6\sigma_w^2$ in our implementation.

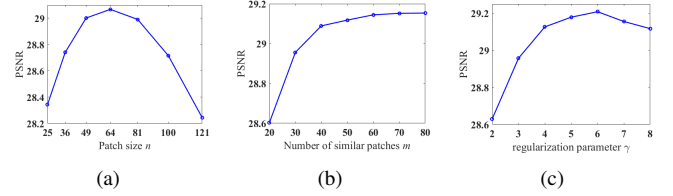


Fig. 8. The PSNR curves as functions of: (a) the patch size n ; (b) the number of similar patches m ; and (c) the regularization parameter γ .

D. Computational complexity

The computational complexity of the proposed method mainly consists of five parts: 1) computing the k -NN nearest

TABLE II
THE AVERAGE PSNR(dB), FSIM (%) AND CORNIA RESULTS OF MIXED NOISE REMOVAL (AWGN+RVIN) ON THE SET OF 12 TEST IMAGES.

Method		TPD [9]	BM3D* [3]	WESNR [13]	SLR [11]	ℓ_1 -NLR	ℓ_0 -NLR	LSM-NLR
$\sigma_w = 10$	$r = 10\%$	28.78	30.57	30.24	30.50	31.25	31.36	32.30
		95.59	97.11	96.65	97.11	97.48	97.57	97.63
		44.54	31.26	31.55	37.88	28.64	26.50	18.75
	$r = 20\%$	27.97	29.46	29.36	29.18	29.46	29.86	30.82
		94.77	96.43	96.06	96.29	96.63	96.78	96.99
		44.68	34.35	33.04	42.48	32.71	30.71	21.41
	$r = 30\%$	27.15	28.30	28.40	27.82	27.74	28.55	29.37
		93.78	95.43	95.25	95.15	95.32	95.78	96.17
		48.28	39.83	37.35	46.08	38.50	33.63	26.07
	$r = 40\%$	26.02	26.67	27.02	26.20	26.74	26.92	27.24
		91.89	93.38	93.69	93.16	93.25	93.93	93.88
		51.08	46.61	40.51	50.37	34.78	39.68	34.24
	$r = 50\%$	24.92	24.54	25.30	24.18	24.72	25.18	25.36
		89.60	89.58	90.99	89.63	90.23	90.96	91.00
		53.30	49.29	45.28	55.49	38.33	41.02	39.29
$\sigma_w = 20$	$r = 10\%$	27.96	27.96	27.69	27.62	28.62	28.90	29.22
		93.76	93.76	92.82	93.66	94.29	94.77	94.78
		51.02	38.15	35.33	38.50	28.42	29.62	25.86
	$r = 20\%$	25.84	27.21	27.09	26.79	27.50	27.78	28.27
		90.49	92.82	92.12	92.58	93.31	93.56	93.80
		52.92	39.68	37.73	41.03	32.19	31.91	27.05
	$r = 30\%$	25.29	26.36	26.42	25.86	26.21	26.74	27.28
		89.43	91.52	91.27	91.18	91.91	92.15	92.67
		53.14	42.22	39.03	43.97	33.79	35.77	30.46
	$r = 40\%$	24.37	24.97	25.24	24.54	24.83	25.37	26.08
		87.13	88.96	89.44	88.73	89.05	89.78	90.83
		56.36	46.90	42.01	45.22	41.19	40.96	30.78
	$r = 50\%$	23.42	23.22	23.86	22.88	23.19	24.00	24.62
		84.70	85.19	87.07	85.14	85.86	86.70	87.92
		56.16	51.58	45.97	49.22	45.22	43.42	38.54
$\sigma_w = 30$	$r = 10\%$	24.58	26.08	26.11	25.60	26.56	26.95	26.98
		86.98	90.11	90.19	90.15	91.06	91.67	91.68
		55.55	43.37	42.54	42.58	34.23	33.62	31.94
	$r = 20\%$	24.21	25.39	25.55	24.85	25.49	25.92	26.20
		85.12	88.81	89.32	88.73	89.66	89.86	90.42
		55.52	44.62	42.54	44.72	36.66	37.22	32.12
	$r = 30\%$	23.76	24.65	24.92	24.03	24.30	25.00	25.39
		84.92	87.33	88.30	87.06	87.91	87.96	89.17
		57.64	47.11	44.72	45.89	38.51	40.50	36.40
	$r = 40\%$	22.95	23.35	23.74	22.83	23.11	23.74	24.33
		82.45	84.48	85.95	84.32	84.47	85.06	86.59
		58.11	50.81	46.64	47.77	47.19	43.90	38.30
	$r = 50\%$	21.99	21.77	22.30	21.35	21.62	22.52	23.18
		79.89	80.73	82.93	80.72	81.13	81.93	83.78
		58.61	53.34	52.31	50.86	51.79	48.84	42.65
$\sigma_w = 50$	$r = 10\%$	22.19	23.66	23.16	22.83	23.83	24.14	24.27
		79.84	84.04	82.71	83.93	85.19	85.54	86.31
		58.92	51.96	49.82	47.74	43.05	39.84	40.34
	$r = 20\%$	21.94	22.99	22.59	22.18	22.73	23.20	23.61
		78.55	82.21	80.92	82.02	83.09	82.66	84.60
		58.21	52.35	50.54	50.08	45.07	46.81	43.75
	$r = 30\%$	21.59	22.29	21.84	21.44	21.59	22.31	22.85
		77.31	80.51	78.94	79.99	80.82	79.86	82.72
		60.18	55.25	52.29	51.61	46.91	49.60	45.71
	$r = 40\%$	20.87	21.05	20.78	20.35	20.73	21.14	21.95
		74.88	77.52	76.17	77.06	76.79	76.39	79.66
		60.06	57.20	54.05	52.22	54.60	52.69	48.37
	$r = 50\%$	19.83	19.62	19.35	19.08	19.34	20.08	20.73
		72.37	74.04	72.51	73.71	73.53	73.74	76.80
		61.50	60.24	55.62	54.67	56.27	54.00	51.71

TABLE III
THE AVERAGE PSNR(dB), FSIM (%) AND CORNIA RESULTS OF MIXED NOISE REMOVAL (AWGN+SPIN+RVIN) ON THE SET OF 12 TEST IMAGES.

Method	BM3D* [3]	WESNR [13]	SLR [11]	ℓ_1 -NLR	ℓ_0 -NLR	LSM-NLR
$\sigma_w = 5, s = 50\%, r = 5\%$	24.39	28.00	27.48	28.76	28.88	29.08
	90.31	94.85	94.94	96.06	96.11	96.46
	44.78	32.88	32.04	28.29	23.31	21.78
$\sigma_w = 10, s = 40\%, r = 10\%$	24.76	27.25	27.27	28.00	28.36	29.20
	89.62	93.53	94.22	95.06	95.14	96.09
	42.99	35.06	37.96	33.33	26.69	23.57
$\sigma_w = 15, s = 30\%, r = 15\%$	24.37	25.98	26.55	27.34	27.40	28.15
	88.50	91.14	92.58	93.32	93.20	94.60
	43.27	36.38	35.53	35.47	33.67	30.59
$\sigma_w = 20, s = 10\%, r = 30\%$	23.63	25.27	25.42	25.74	25.87	26.10
	86.94	90.30	90.32	90.63	90.56	91.32
	54.91	44.14	41.61	40.97	36.18	37.70

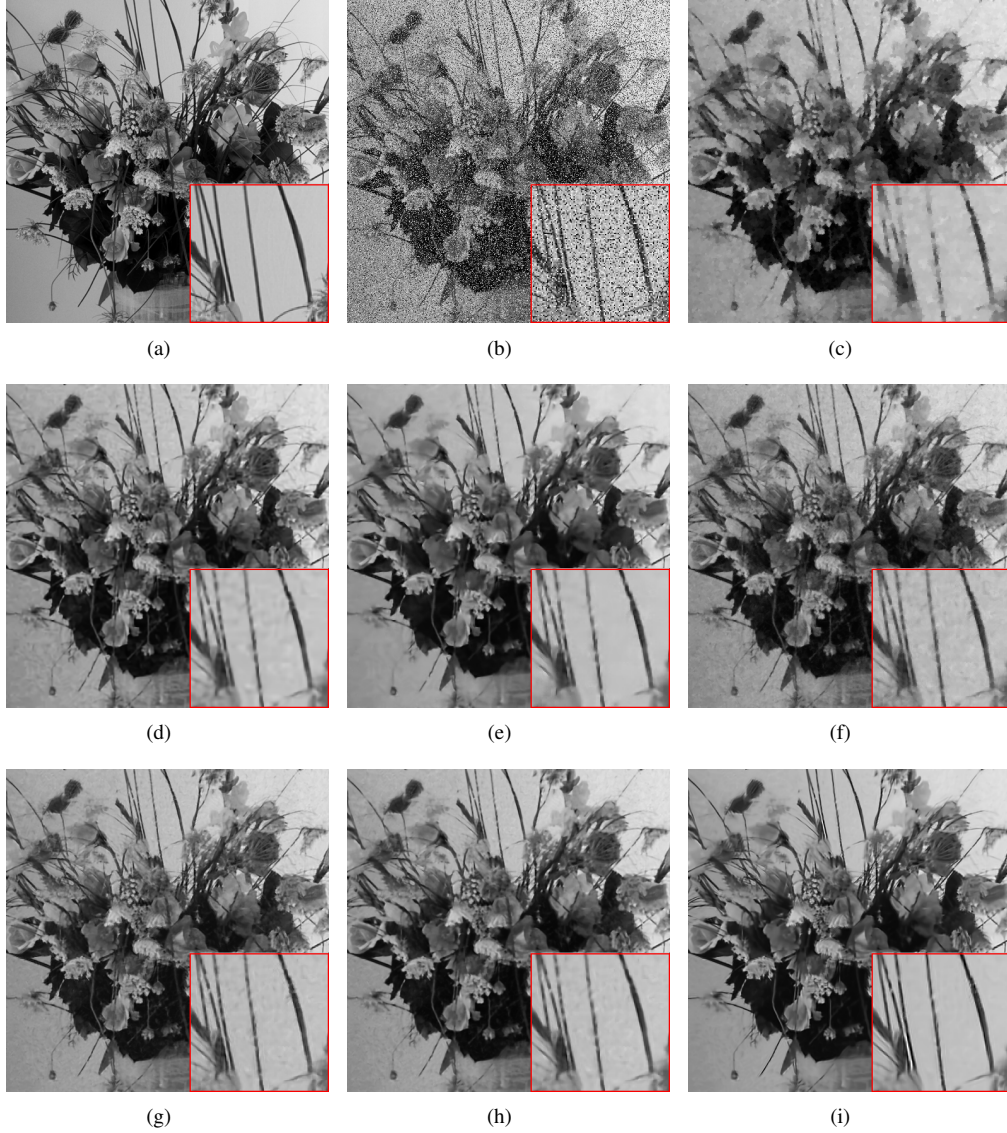


Fig. 5. Denoising results of different methods on test image *Flower* corrupted with AWGN+RVIN($\sigma_w = 20, r = 30\%$). (a) Original image; (b) Noisy image; images denoised by (c) TPD method [9] (PSNR= 22.11dB, FSIM= 0.8707, CORNIA= 49.15); (d) BM3D* [3] (PSNR= 23.22dB, FSIM= 0.9039, CORNIA= 31.26); (e) WESNR [13] (PSNR= 23.04dB, FSIM= 0.8956, CORNIA= 29.95); (f) SLR [11] (PSNR= 23.19dB, FSIM= 0.8989, CORNIA= 32.52); (g) ℓ_1 -NLR (PSNR= 23.68dB, FSIM= 0.9120, CORNIA= 25.50); (h) ℓ_0 -NLR (PSNR= 23.51dB, FSIM= 0.9071, CORNIA= 19.21); (i) proposed **LSM-NLR** (PSNR= **24.36**dB, FSIM= **0.9156**, CORNIA= **14.88**).

TABLE VII
THE AVERAGE PSNR(dB), FSIM (%) AND CORNIA RESULTS OF NOISE REMOVAL (AWGN) ON THE 12 TEST IMAGES.

Method	BM3D [3]	NCSR [6]	SLR [11]	LSM-NLR
$\sigma_w = 10$	34.14	34.16	32.88	34.25
	98.39	98.25	98.12	98.28
	20.91	18.43	25.34	15.04
$\sigma_w = 20$	30.78	30.79	29.07	30.87
	96.25	95.98	95.42	95.99
	28.76	23.57	32.08	19.07
$\sigma_w = 30$	28.91	28.88	26.86	28.94
	94.08	93.51	93.21	93.71
	32.34	28.25	40.48	23.21
$\sigma_w = 50$	26.57	26.30	24.24	26.50
	90.13	89.52	89.19	89.62
	43.96	36.49	43.96	31.07

neighbors per each exemplar patch within a window of size $h \times h$ for patch grouping; 2) solving the σ_w -subproblem;

3) solving the θ and α -subproblem; 4) computing the L_j via singular value thresholding; 5) solving the x -subproblem. The complexity of computing the k -NN patch grouping is $O(T_1 P m (h^2 + \log h^2))$, where T_1 is the number of performed patch grouping, i.e., $T_1 = \lceil T/T_0 \rceil$ wherein $\lceil \cdot \rceil$ denotes the rounding operation, P is the number of extracted similar patches and m is the number of similar patches. The complexity of solving the σ_w -subproblem is $O(TN)$, where N is the number of total pixels of the noisy image. The complexity of solving both the θ and α -subproblems is $O(TN)$. Under the assumption $m \leq n$, the complexity of computing the L_j is $O(TJP(mn^2 + Kmnr_0))$, where n and m denotes the number of rows and column of the matrix L_j , respectively and J denotes the number of ADMM iterations. The complexity of solving the x -subproblem is $O(TJ(Pmn + N))$. The overall complexity of the proposed method under $r_0 \leq m \leq n$ and

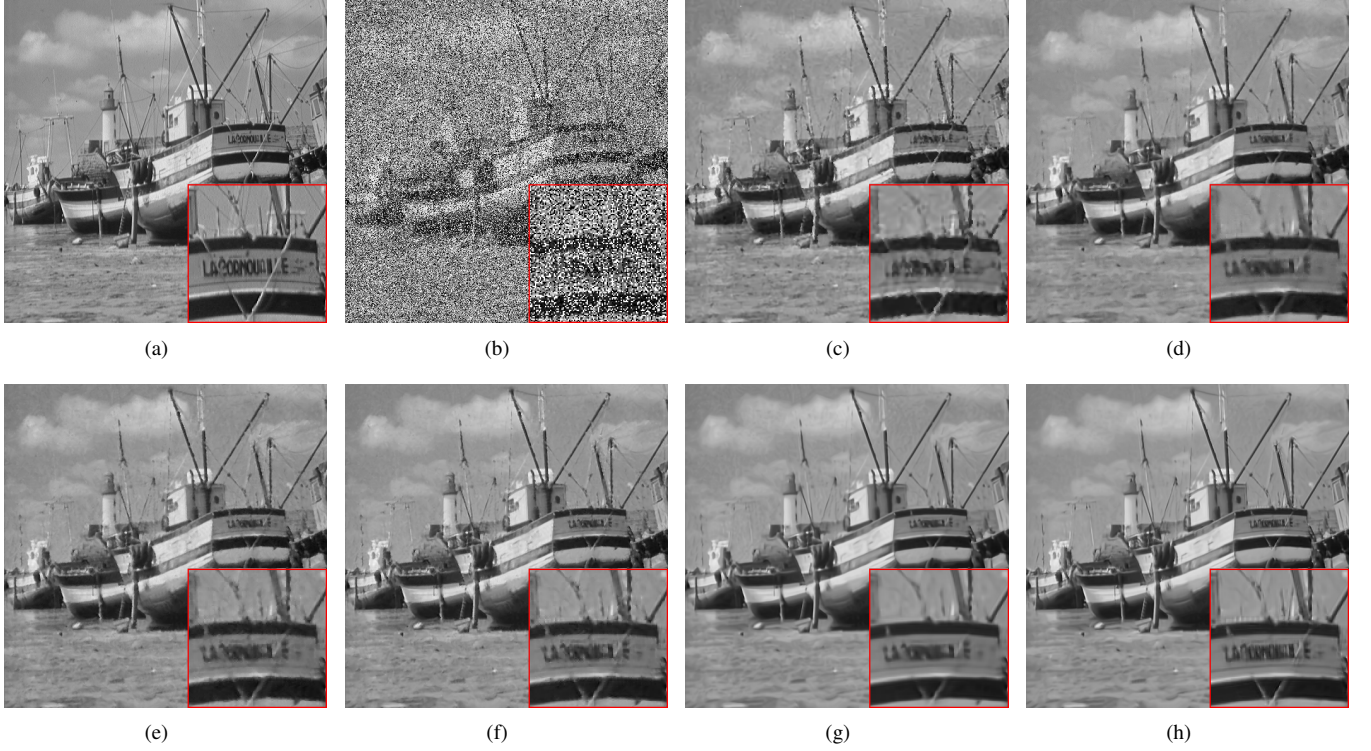


Fig. 6. Denoising results of different methods on test image *Boat* corrupted with AWGN+SPIN+RVIN($\sigma_w = 10$, $s = 40\%$, $r = 10\%$). (a) Original image; (b) Noisy image; images denoised by (c) BM3D* [3] (PSNR= 23.89dB, FSIM= 0.8598, CORNIA = 47.83); (d) WESNR [13] (PSNR= 27.32dB, FSIM= 0.9275, CORNIA= 42.89); (e) SLR [11], (PSNR= 27.35dB, FSIM= 0.9385, CORNIA= 43.95); (f) ℓ_1 -NLR (PSNR= 27.99dB, FSIM= 0.9452, CORNIA= 42.14); (g) ℓ_0 -NLR (PSNR= 27.99dB, FSIM= 0.9396, CORNIA= 41.40); (h) proposed **LSM-NLR** (PSNR= **28.89dB**, FSIM= **0.9482**, CORNIA= **31.47**).

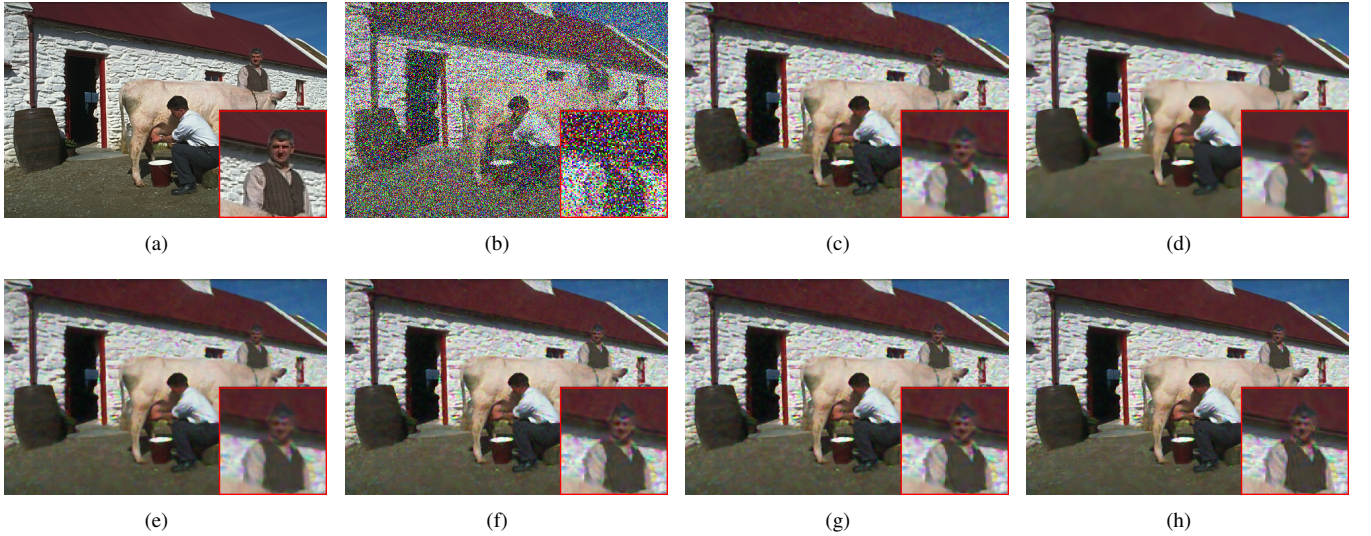


Fig. 7. Denoising results of different methods on test image 385039 of BSD100 dataset corrupted with AWGN+SPIN+RVIN($\sigma_w = 15$, $s = 30\%$, $r = 15\%$). (a) Original image; (b) Noisy image; images denoise by (c) BM3D* [3] (PSNR= 22.57dB, FSIM= 0.8225, CORNIA= 34.45); (d) WESNR [13] (PSNR= 22.63dB, FSIM= 0.7671, CORNIA= 26.18); (e) SLR [11] (PSNR= 22.67dB, FSIM= 0.8034, CORNIA= 31.72); (f) ℓ_1 -NLR (PSNR= 22.88dB, FSIM= 0.8370, CORNIA= 25.85); (g) ℓ_0 -NLR (PSNR= 22.90dB, FSIM= 0.8323, CORNIA= 16.75); (h) proposed **LSM-NLR** (PSNR= **23.43dB**, FSIM= **0.8602**, CORNIA= **11.53**).

$n^2 \leq N$ is

$$O(TJ(Pmn^2 + Kmnr_0 + N) + T_1Pm(h^2 + \log h^2)) \quad (34)$$

The propose algorithm was implemented with matlab language on an Intel core i7-4790 3.6GHz CPU. A running time comparison between the proposed method and other competing

methods on 256×256 test image is shown in Table VIII. From Table VIII, it can be seen that the running time of the proposed LSM-NLR method and ℓ_p -NLR ($p = 0, 1$) methods are comparable, showing that the complexity of LSM modeling is comparable with the ℓ_p ($p = 0, 1$) norm based sparse noise modeling. The proposed method is slightly slower than

TABLE IV
THE AVERAGE PSNR(dB), FSIM (%) AND CORNIA RESULTS OF MIXED NOISE REMOVAL (AWGN+SPIN) ON THE BSD100 DATASET.

Method		BM3D* [3]	WESNR [13]	SLR [11]	ℓ_1 -NLR	ℓ_0 -NLR	LSM-NLR
$\sigma_w = 10$	$s = 10\%$	29.72	29.11	30.68	30.90	31.02	31.48
		91.40	89.63	92.94	93.69	94.25	94.29
		30.63	30.43	29.45	29.45	20.77	24.83
	$s = 20\%$	28.96	28.71	29.89	30.11	30.25	30.78
		91.17	89.43	92.09	92.97	93.69	93.73
		32.80	31.36	27.57	29.95	21.82	22.14
	$s = 30\%$	27.99	28.15	29.04	29.31	29.72	30.21
		90.44	88.81	90.96	92.18	92.77	90.90
		37.09	33.40	28.10	30.38	22.85	19.63
	$s = 40\%$	26.91	27.45	28.14	28.47	28.55	29.24
		89.31	87.84	89.14	91.21	91.57	91.62
		41.84	35.84	29.15	30.35	25.09	18.68
	$s = 50\%$	25.78	26.62	27.17	27.54	27.36	28.17
		87.74	86.45	87.65	89.96	90.06	89.87
		45.09	36.53	29.43	29.94	27.86	19.21
$\sigma_w = 20$	$s = 10\%$	27.48	26.80	27.25	26.77	27.27	29.09
		86.12	82.43	86.78	87.38	88.27	89.25
		37.02	35.01	39.26	41.65	39.25	23.68
	$s = 20\%$	27.06	26.48	26.89	26.61	27.14	28.63
		86.24	82.59	86.25	87.05	88.05	87.75
		39.14	35.86	38.90	40.88	37.74	22.00
	$s = 30\%$	26.45	26.05	26.45	26.44	26.93	28.05
		85.82	82.29	85.32	85.64	85.94	85.97
		42.72	38.60	37.36	39.46	35.40	22.58
	$s = 40\%$	25.68	25.47	25.94	26.22	26.60	27.55
		84.95	81.67	84.19	86.10	86.11	86.13
		47.66	41.01	37.04	37.74	34.44	23.95
	$s = 50\%$	24.79	24.81	25.37	25.86	26.12	26.88
		83.52	80.61	82.77	83.25	83.79	83.83
		51.36	42.37	36.01	35.92	34.05	25.13
$\sigma_w = 30$	$s = 10\%$	26.07	25.43	24.78	24.62	25.22	27.33
		82.75	79.93	81.16	82.98	84.15	84.90
		42.01	40.77	48.54	47.39	44.41	29.24
	$s = 20\%$	25.68	24.94	24.53	24.61	25.27	26.94
		82.84	80.05	81.01	82.79	83.00	83.00
		44.42	43.23	46.78	45.02	42.56	27.89
	$s = 30\%$	25.12	24.35	24.17	24.63	25.24	26.48
		82.41	79.86	80.56	82.56	83.69	84.21
		47.78	45.98	46.11	43.60	40.30	33.45
	$s = 40\%$	24.44	23.70	23.76	24.60	25.11	26.06
		81.44	79.34	79.70	82.14	82.39	82.45
		50.26	48.60	44.18	41.02	38.17	32.30
	$s = 50\%$	23.58	22.92	23.25	24.43	24.81	25.44
		79.89	78.45	78.29	80.43	80.66	80.75
		54.67	49.86	42.33	39.36	37.09	35.62
$\sigma_w = 50$	$s = 10\%$	24.28	19.82	21.15	22.13	22.80	24.05
		78.19	73.45	74.49	76.71	78.05	80.51
		49.60	60.21	58.62	53.20	50.50	48.78
	$s = 20\%$	23.86	17.95	20.87	22.18	22.99	24.70
		78.24	70.34	74.05	76.73	78.28	80.39
		50.31	60.99	56.89	50.54	47.78	43.36
	$s = 30\%$	23.31	16.51	20.54	22.27	23.02	24.70
		77.77	67.83	73.52	76.46	77.78	78.65
		51.87	60.65	55.48	49.23	45.74	41.65
	$s = 40\%$	22.61	15.40	20.15	22.32	22.93	24.41
		76.70	65.75	73.03	75.99	77.00	77.03
		52.63	59.91	52.82	46.02	43.15	40.65
	$s = 50\%$	21.76	14.44	19.70	22.22	22.66	23.96
		74.95	63.90	72.44	75.22	75.60	75.66
		54.23	57.47	49.90	43.29	41.61	41.95

the WESNR [13] and the TPD method [9]. This is because the proposed method use the low-rank approximation based regularization, which is of high computational complexity. However, the proposed method can be speeded up by using fast implementation of low-rank decomposition (e.g., [57]) or using parallel computational technique.

V. CONCLUSION

Removing mixed impulse noise (IN) and additive white Gaussian noise (AWGN) is a challenging problem due to the difficulties in accurately modeling the distributions of the mixed noise. In this paper, we propose an effective mixed IN

and AWGN removal method based on Laplacian Scale Mixture (LSM) modeling and nonlocal low-rank regularization. The impulse noise is modeled with LSM model and both the hidden scale parameters and the IN are jointly estimated. To exploit the nonlocal self-similarity, a non-local regularization is used to regularize the image recovery process. Experimental results show that the proposed method performs significantly better than existing mixture noise removal methods.

REFERENCES

- [1] A. Buades, B. Coll, and J. Morel, "A non-local algorithm for image denoising," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, vol. 2, Jun. 2005, pp. 60–65.

TABLE V
THE AVERAGE PSNR(dB), FSIM (%) AND CORNIA RESULTS OF MIXED NOISE REMOVAL (AWGN+RVIN) ON THE BSD100 DATASET.

Method		BM3D* [3]	WESNR [13]	SLR [11]	ℓ_1 -NLR	ℓ_0 -NLR	LSM-NLR
$\sigma_w = 10$	$r = 10\%$	28.07	27.66	28.33	29.13	29.15	30.12
		88.88	87.25	89.70	91.82	92.51	92.70
		36.44	36.36	36.51	30.55	31.27	24.20
	$r = 20\%$	27.28	27.06	27.38	27.67	27.79	28.54
		87.76	86.12	88.16	90.00	90.38	90.83
		39.86	38.92	41.13	36.11	34.46	27.65
	$r = 30\%$	26.48	26.44	26.41	26.27	26.75	27.27
		86.53	84.91	86.42	87.72	88.20	88.94
		42.47	40.77	45.73	41.11	36.49	31.75
	$r = 40\%$	25.20	25.40	25.13	25.22	25.43	27.27
		82.01	82.60	81.75	81.63	82.61	88.94
		47.58	43.97	50.92	38.23	40.62	34.46
	$r = 50\%$	23.49	24.09	23.49	23.61	23.92	24.37
		79.08	80.20	80.18	79.34	80.36	79.62
		50.31	46.75	55.13	40.87	39.86	36.42
$\sigma_w = 20$	$r = 10\%$	26.12	25.87	26.12	26.97	27.22	27.64
		82.95	79.44	83.82	87.89	87.24	87.44
		43.42	40.63	39.65	30.99	33.73	28.48
	$r = 20\%$	25.58	25.47	25.56	26.07	26.21	26.73
		82.18	78.62	82.75	83.82	85.23	85.38
		45.48	42.33	42.20	33.20	35.67	29.94
	$r = 30\%$	24.95	25.04	24.87	25.03	25.35	25.86
		81.26	77.81	81.44	82.49	82.98	83.35
		47.87	44.23	44.81	36.71	38.32	33.02
	$r = 40\%$	23.85	24.13	23.80	23.85	24.24	24.85
		79.15	46.03	79.70	78.05	80.20	80.44
		51.29	47.18	46.36	45.56	43.20	35.80
	$r = 50\%$	22.36	22.98	22.38	22.43	23.01	23.45
		76.46	74.59	76.20	75.84	76.26	76.83
		54.14	50.50	48.66	48.27	45.89	40.33
$\sigma_w = 30$	$r = 10\%$	24.80	24.86	24.63	25.38	25.71	25.87
		79.20	77.30	80.23	80.96	83.46	82.70
		48.35	45.65	43.37	36.32	34.52	32.93
	$r = 20\%$	24.27	24.46	24.07	24.35	24.65	24.96
		78.59	76.59	79.23	79.44	80.27	80.20
		50.40	46.93	46.54	41.65	41.50	37.64
	$r = 30\%$	23.65	23.97	23.52	23.53	24.01	24.51
		77.75	76.01	77.91	78.46	78.82	78.87
		52.25	49.47	47.09	41.22	43.25	37.50
	$r = 40\%$	22.59	23.03	22.53	22.56	23.01	23.73
		75.70	74.27	75.64	75.05	75.72	75.74
		54.38	51.35	48.66	50.29	47.11	39.27
	$r = 50\%$	21.17	21.74	21.23	21.21	21.89	22.69
		71.34	72.78	71.14	73.04	73.19	71.98
		56.82	54.12	50.76	52.25	50.22	42.58
$\sigma_w = 50$	$r = 10\%$	22.96	22.67	22.34	23.09	23.19	23.40
		74.94	74.93	75.93	76.17	76.15	76.19
		55.44	51.37	49.64	46.47	44.97	44.71
	$r = 20\%$	22.42	22.21	22.08	22.38	22.60	23.08
		73.95	73.78	74.73	74.19	74.38	74.88
		56.94	53.47	50.55	46.49	47.12	44.42
	$r = 30\%$	21.74	21.47	21.50	21.36	21.79	22.48
		73.26	72.65	73.53	73.77	73.51	73.34
		57.99	54.96	51.94	48.79	50.62	46.64
	$r = 40\%$	20.67	20.50	20.58	20.58	20.82	21.85
		69.41	70.52	71.27	70.27	70.34	70.00
		59.38	56.23	53.04	55.08	53.39	48.76
	$r = 50\%$	19.33	19.13	19.25	19.14	19.65	20.62
		67.47	67.97	69.40	69.01	69.13	68.66
		60.98	56.38	55.21	56.45	56.68	54.94

TABLE VI
THE AVERAGE PSNR(dB), FSIM (%) AND CORNIA RESULTS OF MIXED NOISE REMOVAL (AWGN+SPIN+RVIN) ON THE BSD100 DATASET.

Method	BM3D* [3]	WESNR [13]	SLR [11]	ℓ_1 -NLR	ℓ_0 -NLR	LSM-NLR
$\sigma_w = 5, s = 50\%, r = 5\%$	24.40	26.72	25.77	26.58	26.65	26.76
	86.97	86.28	82.96	88.29	88.73	88.77
	34.61	33.37	36.37	31.03	26.11	24.05
$\sigma_w = 10, s = 40\%, r = 10\%$	24.40	26.20	25.87	26.13	26.45	27.11
	85.36	86.23	85.04	86.60	86.09	88.44
	43.01	38.97	39.22	36.41	28.50	26.59
$\sigma_w = 15, s = 30\%, r = 15\%$	24.38	25.61	25.32	25.66	25.70	26.34
	83.81	82.14	82.25	84.33	84.09	86.31
	48.16	37.55	40.33	38.36	35.48	32.24
$\sigma_w = 20, s = 10\%, r = 30\%$	22.88	23.91	24.27	24.58	24.55	24.70
	79.50	80.65	79.24	80.63	80.27	79.72
	53.67	45.56	45.79	42.90	42.82	41.76

TABLE VIII
RUNNING TIME (SECOND) COMPARISON ON A 256×256 TEST IMAGE.

Method	TPD [9]	BM3D* [3]	WESNR [13]	SLR [11]	ℓ_1 -NLR	ℓ_0 -NLR	LSM-NLR
$\sigma_w = 30, s = 30\%$	17	0.6	18	32	23	20	12
$\sigma_w = 30, s = 50\%$	20	0.6	18	30	24	20	12
$\sigma_w = 30, r = 30\%$	14	0.6	18	19	19	19	60
$\sigma_w = 30, r = 50\%$	16	0.6	18	19	19	37	88
$\sigma_w = 5, r = 5\%, s = 50\%$	–	0.6	12	25	30	55	47
$\sigma_w = 10, r = 10\%, s = 40\%$	–	0.6	12	28	29	37	32
$\sigma_w = 15, r = 15\%, s = 30\%$	–	0.6	16	26	22	37	31

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Tao Huang is currently pursuing the Ph.D. degree in Intelligent Information Processing at Xidian University, Xi'an, China. His research interests include image processing, Sparse representation, and low rank matrix recovery.



Weisheng Dong (M'11) received the B.S. degree in electronic engineering from the Huazhong University of Science and Technology, Wuhan, China, in 2004, and the Ph.D. degree in circuits and system from Xidian University, Xi'an, China, in 2010. He was a Visiting Student with Microsoft Research Asia, Beijing, China, in 2006. From 2009 to 2010, he was a Research Assistant with the Department of Computing, Hong Kong Polytechnic University, Hong Kong. In 2010, he joined the School of Electronic Engineering, Xidian University, as a Lecturer,

where he has been a Professor since 2016. His research interests include inverse problems in image processing, sparse signal representation, and image compression.

He was a recipient of the Best Paper Award at the SPIE Visual Communication and Image Processing (VCIP) in 2010. He is currently serving as an associate editor of IEEE Transactions on Image Processing.



Xuemei Xie received M.S. degree in electronic engineering from Xidian University in 1994 and Ph.D. degree in electrical and electronic engineering from the University of Hong Kong in 2004. Her research interests are digital signal processing, multi-rate filter banks, and wavelet transforms.



Guangming Shi (SM'10) received the B.S. degree in Automatic Control in 1985, the M.S. degree in Computer Control and Ph.D. degree in Electronic Information Technology, all from Xidian University in 1988 and 2002, respectively.

He joined the School of Electronic Engineering, Xidian University, in 1988. From 1994 to 1996, as a Research Assistant, he cooperated with the Department of Electronic Engineering at the University of Hong Kong. Since 2003, he has been a Professor in the School of Electronic Engineering at Xidian University, and in 2004 the head of National Instruction Base of Electrician and Electronic (NIBEE). From June to December in 2004, he had studied in the Department of Electronic Engineering at University of Illinois at Urbana-Champaign (UIUC). Presently, he is the Deputy Director of the School of Electronic Engineering, Xidian University, and the academic leader in the subject of Circuits and Systems. His research interests include compressed sensing, theory and design of multirate filter banks, image denoising, low-bit-rate image/video coding and implementation of algorithms for intelligent signal processing (using DSP and FPGA). He has authored or co-authored over 60 research papers.



Xiang Bai (S'07-M'09-SM'14) received the B.S., M.S., and Ph.D. degrees in electronics and information engineering from the Huazhong University of Science and Technology (HUST), Wuhan, China, in 2003, 2005, and 2009, respectively. He is currently a Professor with the School of Electronic Information and Communications, HUST. He is also the Vice-Director of the National Center of Anti-Counterfeiting Technology, HUST. His research interests include object recognition, shape analysis, scene text recognition, and intelligent systems.